GATE ME 90 Days Challenge

Day 44/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Heat Transfer	Convection	60 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on Newton's law of cooling, the dimensionless numbers (Re, Pr, Nu, Gr, Ra, St), velocity and thermal boundary layers, laminar and turbulent flat-plate correlations, tube flow (entry length, Dittus–Boelter, fully developed Nu), free convection on vertical plates and the Reynolds–Colburn analogy. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The Nusselt number hL/k at a surface represents the ratio of

(A) heat conducted in the solid to heat convected by the fluid

(B) convective heat transfer to the heat that would be conducted across a stagnant fluid layer of thickness L

(C) momentum diffusivity to thermal diffusivity

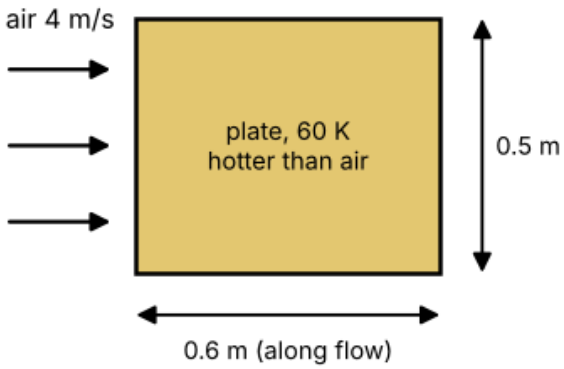
(D) buoyancy force to viscous force

Q2 NAT

Engine oil ($\nu = 1.0 \times 10^{-4}$ m²/s, Pr = 500) flows at 0.5 m/s through a 25 mm diameter tube. Using $L_h \approx 0.05\ Re\ D$ for the hydrodynamic entry length and $L_t \approx L_h\ Pr$ for the thermal entry length, the thermal entry length (in m, round off to one decimal place) is _____

Q3 NAT

Air at 4 m/s flows parallel to the 0.6 m side of a flat plate 0.6 m × 0.5 m, as shown. The plate is 60 °C hotter than the air. Air properties: $\nu = 1.5 \times 10^{-5}$ m²/s, k = 0.026 W/m·K, Pr = 0.71. Using $Nu_L = 0.664\ Re_L^{1/2}\ Pr^{1/3}$, the heat transfer rate from the top face (in W, round off to one decimal place) is _____



Q4 MCQ

The thermal boundary layer on a heated flat plate is thicker than the velocity boundary layer when

(A) Pr > 1
(B) Pr = 1
(C) the flow is turbulent
(D) Pr < 1

Q5 NAT

Air (k = 0.027 W/m·K, $\nu = 1.6 \times 10^{-5}$ m²/s, Pr = 0.7) flows at 10 m/s through a 50 mm diameter duct and is being cooled. Using the Dittus–Boelter equation $Nu = 0.023\ Re^{0.8}\ Pr^{n}$, the heat transfer coefficient (in W/m²·K, round off to one decimal place) is _____

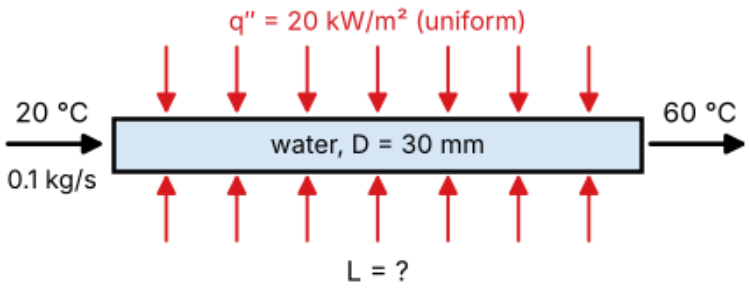
Q6 MCQ

For hydrodynamically and thermally fully developed laminar flow in a circular tube with a uniform wall temperature, the Nusselt number based on diameter is

(A) 4.36
(B) 3.66
(C) 5.39
(D) It increases with Reynolds number

Q7 NAT

Water (c_p = 4180 J/kg·K) at 0.1 kg/s is heated from 20 °C to 60 °C in a 30 mm diameter tube whose wall supplies a uniform heat flux of 20 kW/m², as shown. The required tube length (in m, round off to two decimal places) is _____



Q8 MCQ

The Grashof number for a vertical plate of height L at temperature T_s in a fluid at T_∞ is

(A) $g\beta(T_s - T_\infty)L^3/\nu^2$
(B) $g\beta(T_s - T_\infty)L^2/\nu$
(C) $\rho uL/\mu$
(D) $g\beta(T_s - T_\infty)L^3/(\nu\alpha)$

Q9 NAT

A vertical plate 2 m high is at 100 °C in still air at 20 °C. Air properties at the film temperature: $\nu = 1.9 \times 10^{-5}$ m²/s, k = 0.029 W/m·K, Pr = 0.7, $\beta = 1/333$ K^{-1}; g = 9.81 m/s². Using $Nu_L = 0.10\ Ra_L^{1/3}$ for turbulent flow, the average heat transfer coefficient (in W/m²·K, round off to two decimal places) is _____

Q10 MCQ

For laminar flow over a flat plate, if the free-stream velocity is doubled (all else unchanged), the average heat transfer coefficient becomes

(A) 2 times
(B) 4 times
(C) $\sqrt{2}$ times
(D) $2^{0.8}$ times

Q11 NAT

Air ($\nu = 1.6 \times 10^{-5}$ m²/s, k = 0.027 W/m·K, Pr = 0.7) flows at 30 m/s over a flat plate 2 m long whose boundary layer is tripped so that it is turbulent from the leading edge. Using $Nu_L = 0.037\ Re_L^{0.8}\ Pr^{1/3}$, the average heat transfer coefficient (in W/m²·K, round off to one decimal place) is _____

Q12 MCQ

The velocity and thermal boundary layers on a flat plate have the same thickness when

(A) Pr = 1
(B) Re = 1
(C) Nu = 1
(D) $Gr = Re^2$

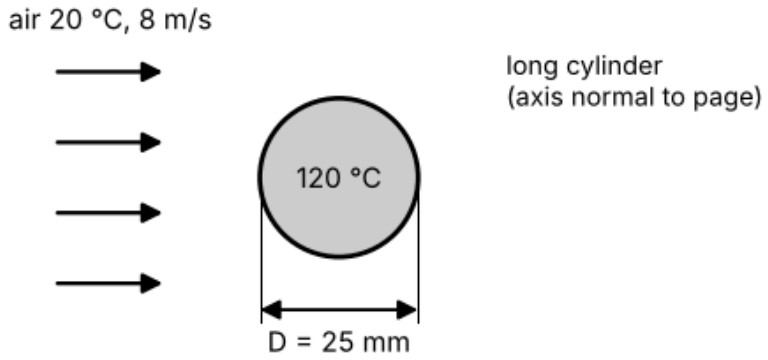
Q13 MCQ

In laminar flow through a tube heated from the entrance, the local heat transfer coefficient along the tube

(A) is zero at the entrance and rises to a constant value
(B) is constant along the whole tube
(C) is highest near the entrance and decreases to a constant fully developed value
(D) rises continuously along the tube

Q14 NAT

Air at 20 °C flows at 8 m/s across a long cylinder of 25 mm diameter whose surface is at 120 °C, as shown. Air properties: $\nu = 1.6 \times 10^{-5}$ m²/s, k = 0.027 W/m·K, Pr = 0.7. Using $Nu_D = 0.193\ Re_D^{0.618}\ Pr^{1/3}$, the heat loss per metre length of the cylinder (in W/m, round off to the nearest integer) is _____



Q15 MCQ

A metal ball is cooled in an air stream. In calculating its Biot number and the Nusselt number of the air flow, the thermal conductivity used is

(A) that of the metal in both

(B) that of the air in both

(C) that of the metal for Bi and that of the air for Nu

(D) that of the air for Bi and that of the metal for Nu

Answer Key

Q	Type	Answer
Q1	MCQ	(B) convective heat transfer to the heat that would be conducted across a stagnant fluid layer of thickness L
Q2	NAT	78.1 (accept 77.3 to 78.9)
Q3	NAT	184.8 (accept 183.0 to 186.6)
Q4	MCQ	(D) Pr < 1
Q5	NAT	44.0 (accept 43.6 to 44.4)
Q6	MCQ	(B) 3.66
Q7	NAT	8.87 (accept 8.78 to 8.96)
Q8	MCQ	(A) $g\beta(T_s - T_\infty)L^3/\nu^2$
Q9	NAT	4.81 (accept 4.77 to 4.86)
Q10	MCQ	(C) $\sqrt{2}$ times
Q11	NAT	80.6 (accept 79.8 to 81.4)
Q12	MCQ	(A) Pr = 1
Q13	MCQ	(C) is highest near the entrance and decreases to a constant fully developed value
Q14	NAT	495 (accept 490 to 500)
Q15	MCQ	(C) that of the metal for Bi and that of the air for Nu

Detailed Solutions

Q1 **Answer: (B) convective heat transfer to the heat that would be conducted across a stagnant fluid layer of thickness L**

Concept: $Nu = hL/k_f = (h\Delta T)/(k_f \Delta T/L)$: convective heat flux divided by the pure-conduction flux through the same fluid layer.

Nu = 1 means the fluid is effectively stagnant (pure conduction); larger Nu means stronger convection.

Option (A) describes the Biot number, which uses k of the *solid*; (C) is Pr and (D) is Gr.

Exam tip: Nu and Bi look alike (hL/k). Nu uses the fluid's k, Bi uses the solid's k.

Q2 **Answer: 78.1 (accept 77.3 to 78.9)**

Concept: In laminar flow the entry lengths grow with Re; the thermal one is longer by the factor Pr.

$Re = uD/\nu = 0.5 \times 0.025/10^{-4} = 125$ (laminar)

$L_h = 0.05 \times 125 \times 0.025 = 0.156$ m

$L_t = 0.156 \times 500 =$ **78.1 m**

Exam tip: For oils (Pr ≫ 1) the flow is thermally developing over almost any practical tube, so fully developed Nu = 3.66 underestimates h.

Q3 **Answer: 184.8 (accept 183.0 to 186.6)**

Concept: Laminar flat plate ($Re_L < 5 \times 10^5$): average $Nu = 0.664\, Re_L^{1/2}\, Pr^{1/3}$, with L measured along the flow.

$Re_L = 4 \times 0.6/1.5 \times 10^{-5} = 1.6 \times 10^5$ (laminar)

$Nu = 0.664 \times 400 \times 0.892 = 236.9$; $h = 236.9 \times 0.026/0.6 = 10.27$ W/m²·K

$Q = hA\Delta T = 10.27 \times (0.6 \times 0.5) \times 60 =$ **184.8 W**

Exam tip: The average h over 0 to L is twice the local h at x = L for laminar flow.

Q4 **Answer: (D) Pr < 1**

Concept: $\delta/\delta_t \approx Pr^{1/3}$. Pr = ν/α compares how fast momentum and heat diffuse.

If Pr < 1 (gases ≈ 0.7, liquid metals ≈ 0.01), heat diffuses faster than momentum, so $\delta_t > \delta$.

For oils and water (Pr > 1) the thermal layer is thinner.

Exam tip: Pr = 1 makes the two profiles identical: this is the basis of the Reynolds analogy.

Q5 **Answer: 44.0 (accept 43.6 to 44.4)**

Concept: Dittus–Boelter for fully developed turbulent flow ($Re > 10^4$): n = 0.4 when the fluid is heated, n = 0.3 when it is cooled.

$Re = 10 \times 0.05/1.6 \times 10^{-5} = 31\,250$ (turbulent)

$Nu = 0.023 \times 31250^{0.8} \times 0.7^{0.3} = 0.023 \times 3943 \times 0.899 = 81.5$

$h = 81.5 \times 0.027/0.05 =$ **44.0 W/m²·K**

Exam tip: In turbulent tube flow $h \propto u^{0.8}/D^{0.2}$, so doubling the velocity raises h by $2^{0.8} = 1.74$.

Q6 **Answer: (B) 3.66**

Concept: In fully developed laminar flow both velocity and dimensionless temperature profiles stop changing, so Nu is a constant.

Uniform wall temperature: Nu = **3.66**; uniform wall heat flux: Nu = 4.36.

Nu does not depend on Re or Pr once the flow is fully developed (laminar).

Exam tip: Since Nu is fixed, h = 3.66k/D: a smaller tube gives a larger h.

Q7 **Answer: 8.87 (accept 8.78 to 8.96)**

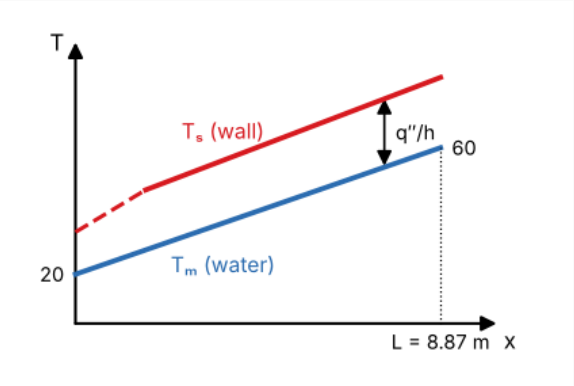
Concept: Energy balance on the fluid: $q''(\pi DL) = \dot{m}c_p(T_o - T_i)$. With uniform q'' the mean temperature rises linearly along the tube.

Heat needed: $Q = 0.1 \times 4180 \times 40 = 16\,720$ W

Surface needed: $\pi DL = 16720/20000 = 0.836$ m²

$L = 0.836/(\pi \times 0.03) =$ **8.87 m**

Exam tip: With uniform heat flux, $T_s - T_m = q''/h$ stays constant in the fully developed region, so both lines are parallel.



Q8 **Answer: (A) $g\beta(T_s - T_\infty)L^3/\nu^2$**

Concept: Gr = buoyancy force/viscous force = $g\beta\Delta TL^3/\nu^2$.

Option (D) is the Rayleigh number Ra = Gr Pr; option (C) is the Reynolds number.

For an ideal gas $\beta = 1/T_f$, with T_f the absolute film temperature.

Exam tip: Laminar-to-turbulent transition for a vertical plate occurs at Ra ≈ 10^9.

Q9 **Answer: 4.81 (accept 4.77 to 4.86)**

Concept: Free convection on a vertical plate: $Ra = g\beta\Delta TL^3Pr/\nu^2$; turbulent when Ra > 10^9.

$Ra = 9.81 \times (80/333) \times 2^3 \times 0.7/(1.9 \times 10^{-5})^2 = 3.66 \times 10^{10}$ (turbulent)

$Nu = 0.10 \times (3.66 \times 10^{10})^{1/3} = 331.9$

$h = 331.9 \times 0.029/2 =$ **4.81 W/m²·K**

Exam tip: With the 1/3 power, L cancels: h in turbulent free convection does not depend on plate height.

Q10 **Answer: (C) √2 times**

Concept: Laminar: $Nu_L = 0.664\,Re_L^{1/2}\,Pr^{1/3}$, so $h \propto u^{1/2}$.

Doubling u multiplies h by $2^{1/2} =$ **1.414**.

Option (D) would apply to turbulent flow ($h \propto u^{0.8}$).

Exam tip: Remember the exponents: laminar 1/2, turbulent 0.8, natural laminar $Ra^{1/4}$, natural turbulent $Ra^{1/3}$.

Q11 **Answer: 80.6 (accept 79.8 to 81.4)**

Concept: For a fully turbulent boundary layer on a plate, local $Nu_x = 0.0296\ Re_x^{0.8}\ Pr^{1/3}$; averaging over L gives $0.037\ Re_L^{0.8}\ Pr^{1/3}$.

$Re_L = 30 \times 2/1.6 \times 10^{-5} = 3.75 \times 10^6$

$Nu = 0.037 \times (3.75 \times 10^6)^{0.8} \times 0.888 = 5968$

$h = 5968 \times 0.027/2 =$ **80.6 W/m²·K**

Exam tip: If the first part of the plate is laminar, use the mixed form $Nu = (0.037\ Re^{0.8} - 871)\ Pr^{1/3}$.

Q12 **Answer: (A) Pr = 1**

Concept: $\delta/\delta_t = Pr^{1/3}$; the momentum and energy equations become identical in form when $\nu = \alpha$.

So $\delta = \delta_t$ exactly when **Pr = 1**.

This similarity is what gives the Reynolds analogy $St = C_f/2$.

Exam tip: Gases (Pr ≈ 0.7) come close, which is why the Reynolds/Colburn analogy works well for air.

Q13 **Answer: (C) is highest near the entrance and decreases to a constant fully developed value**

Concept: At the entrance the thermal boundary layer is very thin, so the temperature gradient at the wall (and h) is large.

As the thermal layer grows, h falls and settles at the fully developed value (Nu = 3.66 or 4.36).

So the average h of a short tube is higher than the fully developed value.

Exam tip: The same 'high near the start' behaviour is seen on a flat plate, where $h_x \propto x^{-1/2}$.

Q14 **Answer: 495 (accept 490 to 500)**

Concept: Cross-flow over a cylinder (Hilpert type): $Nu_D = C\ Re_D^{m}\ Pr^{1/3}$, with C and m chosen for the Re range (here 4000 to 40 000).

$Re_D = 8 \times 0.025/1.6 \times 10^{-5} = 12\ 500$

$Nu = 0.193 \times 12500^{0.618} \times 0.888 = 58.3$; $h = 58.3 \times 0.027/0.025 = 63.0$ W/m²·K

$Q' = h(\pi D)\Delta T = 63.0 \times \pi \times 0.025 \times 100 =$ **495 W/m**

Exam tip: The characteristic length for a cylinder in cross-flow is its diameter, not its length.

Q15 **Answer: (C) that of the metal for Bi and that of the air for Nu**

Concept: $Bi = hL_c/k_{solid}$ compares conduction inside the solid with convection at its surface.

$Nu = hD/k_{fluid}$ is the dimensionless temperature gradient in the fluid at the wall.

So k_{metal} goes into Bi and k_{air} into Nu: option **(C)**.

Exam tip: For a sphere, Bi uses $L_c = D/6$ (lumped) while Nu uses D.

What next?

Day 44 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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